

Tutorial 8: Memo

Question 9.2

Using the wave:

$$i_s = 15 \cos(25\pi t + 25^\circ) A$$

Find:

01	Amplitude [1]
02	Angular frequency [1]
03	Frequency [1]
04	Current at 2 ms [1]

Memo

01) $Amplitude = 15 A$

02) $\omega = 25\pi \text{ rad/s}$

03) $\omega = 2\pi f$

$$f = \frac{\omega}{2\pi}$$

$$= \frac{25\pi}{2\pi}$$

$$= 12.5 \text{ Hz}$$

04) Convert either 25° to radians or 25π to degrees

$$i_s(t) = 15 \cos(25\pi t + 25^\circ)$$

$$i_s(2ms) = 15 \cos\left(25\pi \left(\frac{180}{\pi}\right) t + 25^\circ\right)$$

$$i_s(2ms) = 12.4356 A$$

Question 9.6

Find which wave leads and give the magnitude.

05) $v(t) = 10 \cos(4t - 60^\circ)V$

$i(t) = 4 \sin(4t + 50^\circ)A$

06) $v_1(t) = 4 \cos(377t + 10^\circ)V$

$v_2(t) = -20 \cos(377t)V$

07) $x(t) = 13 \cos(2t) + 5 \sin(2t)$

$y(t) = 15 \sin(2t - 11.8^\circ)$

Memo

05) $v(t) = 10 \cos(4t - 60^\circ)$

$i(t) = 4 \sin(4t + 50^\circ)$

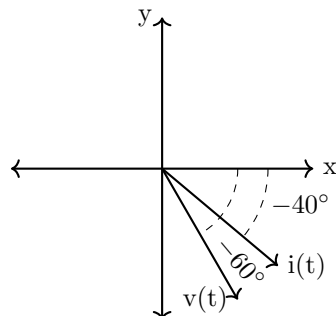
$= 4 \cos(4t + 50^\circ - 90^\circ)$

$= 4 \cos(4t - 40^\circ)$

Convert the sin to cos

$i(t) = 4 \cos(4t + 50^\circ - 90^\circ)$

$= 4 \cos(4t - 40^\circ)$



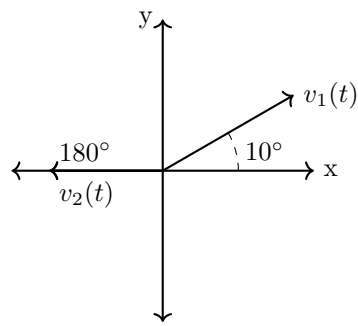
Therefore $i(t)$ leads $v(t)$ by 20°

06) $v_1(t) = 4 \cos(377t + 10^\circ)$

$v_2(t) = -20 \cos(377t)$

Convert the -cos to cos

$v_2(t) = 20 \cos(377t + 180^\circ)$



Therefore $v_2(t)$ leads $v_1(t)$ by 170°

07) Convert the sin to cos

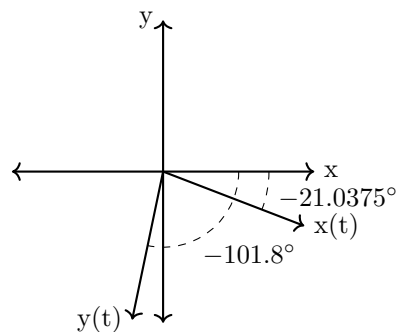
$$\begin{aligned} x(t) &= 13 \cos(2t) + 5 \sin(2t) \\ &= 13 \cos(2t) + 5 \cos(2t - 90^\circ) \end{aligned}$$

$$\begin{aligned} y(t) &= 15 \sin(2t - 11.8^\circ) \\ &= 15 \cos(2t - 11.8^\circ - 90^\circ) \\ &= 15 \cos(2t - 101.8^\circ) \end{aligned}$$

Convert to phasor

$$\begin{aligned} x(t) &= 13 \cos(2t) + 5 \cos(2t - 90^\circ) \\ &= 13 \angle 0^\circ + 5 \angle -90^\circ \\ &= 13.928 \angle -21.0375^\circ \end{aligned}$$

$$\begin{aligned} y(t) &= 15 \sin(2t - 101.8^\circ) \\ &= 15 \angle -101.8^\circ \end{aligned}$$



Therefore $x(t)$ leads $y(t)$ by 80.76°

Question 9.13	
Simplify the following phasors.	
08)	$\frac{2 + j3}{1 - j6} + \frac{7 - j8}{-5 + j11}$
09)	$\frac{(5\angle 10^\circ)(10\angle -40^\circ)}{(4\angle -80^\circ)(-6\angle 50^\circ)}$
10)	$\begin{vmatrix} 2 + j3 & -j2 \\ -j2 & 8 - j5 \end{vmatrix}$
Memo	
08)	$\frac{2 + j3}{1 - j6} + \frac{7 - j8}{-5 + j11}$ $= -1.275 + j0.152$
09)	$\frac{(5\angle 10^\circ)(10\angle -40^\circ)}{(4\angle -80^\circ)(-6\angle 50^\circ)}$ $= \frac{50\angle -30^\circ}{24\angle -210^\circ}$ $= 2.083\angle 180^\circ$
10)	$\begin{vmatrix} 2 + j3 & -j2 \\ -j2 & 8 - j5 \end{vmatrix}$ $= (2 + j3)(8 - j5) - (-j2)(-j2)$ $= 35 + j14$

Question 9.19	
Simplify the following:	
11)	$3 \cos(20t + 10^\circ) - 5 \cos(20t - 30^\circ)$
12)	$40 \sin(50t) + 30 \cos(50t - 45^\circ)$
13)	$20 \sin(400t) + 10 \cos(400t + 60^\circ) - 5 \sin(400t - 20^\circ)$
Memo 11) $3 \cos(20t + 10^\circ) - 5 \cos(20t - 30^\circ)$ $= 3\angle 10^\circ - 5\angle -30^\circ$ $= 3.319\angle 114.484^\circ$ $= 3.319 \cos(20t + 114.5^\circ)$ 12) $40 \sin(50t) + 30 \cos(50t - 45^\circ)$ $= 40 \cos(50t - 90^\circ) + 30 \cos(50t - 45^\circ)$ $= 40\angle -90^\circ + 30\angle -45^\circ$ $= 64.785\angle -70.886^\circ$ $= 64.785 \cos(50t - 70.886^\circ)$ 13) $20 \sin(400t) + 10 \cos(400t + 60^\circ) - 5 \sin(400t - 20^\circ)$ $= 20 \cos(400t - 90^\circ) + 10 \cos(400t + 60^\circ) - 5 \cos(400t - 20^\circ - 90^\circ)$ $= 20 \cos(400t - 90^\circ) + 10 \cos(400t + 60^\circ) - 5 \cos(400t - 110^\circ)$ $= 20\angle -90^\circ + 10\angle 60^\circ - 5\angle -110^\circ$ $= 9.44\angle -44.705^\circ$ $= 9.44 \cos(400t - 44.705^\circ)$	

Question 9.20

Find $\mathbf{Z}$ if:

$$i = 7.5 \cos(10t + 30^\circ)$$

$$v = 120 \cos(10t + 75^\circ)$$

Memo

$$i = 7.5 \cos(10t + 30^\circ) = 7.5 \angle 30^\circ$$

$$v = 120 \cos(10t + 75^\circ) = 120 \angle 75^\circ$$

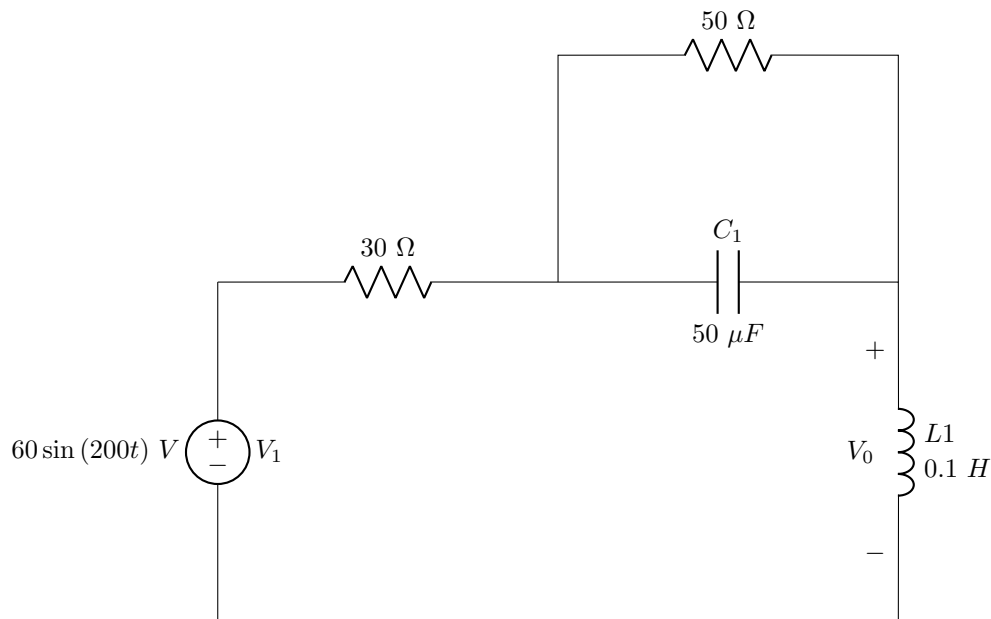
$$Z = \frac{V}{I}$$

$$= \frac{120 \angle 75^\circ}{7.5 \angle 30^\circ}$$

$$= 16 \angle 45^\circ \Omega$$

$$= 11.314 + j11.314 \Omega$$

Question 9.42

Find V_0 :

Memo

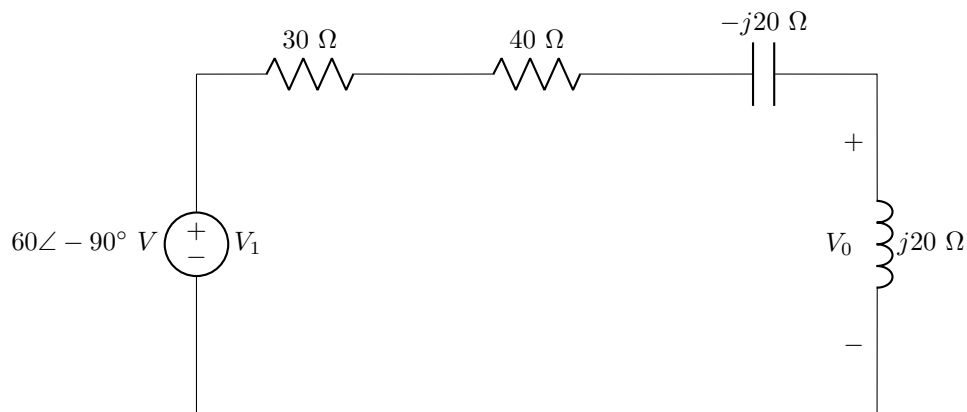
$$\begin{aligned}
 V_1 &= 60 \sin(200t) \\
 &= 60 \cos(200t - 90^\circ) \\
 &= 60 \angle -90^\circ
 \end{aligned}$$

$$\begin{aligned}
 Z_{C_1} &= \frac{1}{j\omega C_1} \\
 &= \frac{1}{j(200)(50 \times 10^{-6})} \\
 &= -j100 \, \Omega
 \end{aligned}$$

$$\begin{aligned}
 Z_{L_1} &= j\omega L \\
 &= j(200)(0.1) \\
 &= j20 \, \Omega
 \end{aligned}$$

The $50 \, \Omega$ and C_1 capacitor are in parallel:

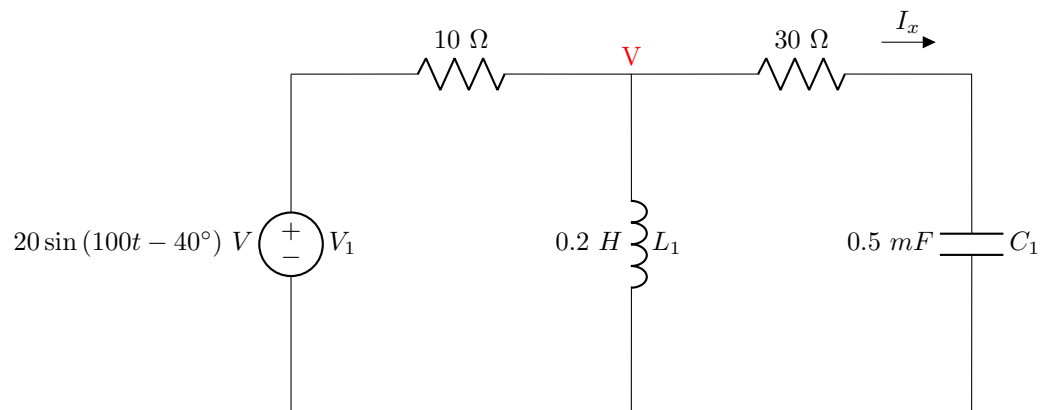
$$\begin{aligned}
 50 || Z_{C_1} &= \frac{50(-j100)}{50 - j100} \\
 &= 40 - j20
 \end{aligned}$$



Use voltage division to find V_0 :

$$\begin{aligned} V_0 &= V_1 \frac{j20}{30 + 40 - j20 + j20} \\ &= \frac{(60\angle -90^\circ)(j20)}{30 + 40 - j20 + j20} \\ &= 17.143\angle 0^\circ \text{ V} \\ &= 17.143 \cos(200t) \text{ V} \end{aligned}$$

Question 9.48

Find $\mathbf{I}_x$:

Memo

$$\begin{aligned}
 V_1 &= 20 \sin(100t - 40^\circ) \\
 &= 20 \cos(100t - 40^\circ - 90^\circ) \\
 &= 20 \sin(100t - 130^\circ) \\
 &= 20 \angle -130^\circ
 \end{aligned}$$

$$\begin{aligned}
 Z_{L_1} &= j\omega L_1 \\
 &= j(100)(0.2) \\
 &= j20
 \end{aligned}$$

$$\begin{aligned}
 Z_{C_1} &= \frac{1}{j\omega C_1} \\
 &= \frac{1}{j(100)(0.5 \times 10^{-3})} \\
 &= -j20
 \end{aligned}$$

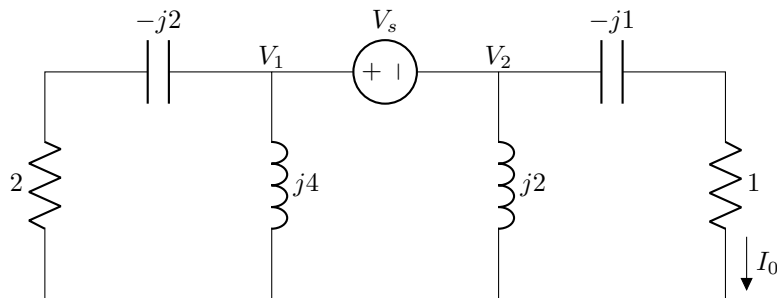
Use nodal analysis at top center node (V):

$$\frac{V - 20 \angle -130^\circ}{10} + \frac{V}{j20} + \frac{V}{30 - j20} = 0$$

$$V = -6.435 - j14.258 \text{ V}$$

$$\begin{aligned}
 I_x &= \frac{V}{30 - j20} \\
 &= \frac{-6.435 - j14.258}{30 - j20} \\
 &= 0.43386 \angle -80.6^\circ \text{ A} \\
 &= 0.43386 \cos(100t - 80.6^\circ) \text{ A}
 \end{aligned}$$

Question 9.54

Find $\mathbf{V}_s$ if $\mathbf{I}_0 = 2 \text{ A}$:

Memo

Supernode:

$$V_s = V_1 - V_2$$

$$I_0 = \frac{V_2}{1 - j1}$$

$$\begin{aligned} V_2 &= I_0(1 - j1) \\ &= 2(1 - j1) \\ &= 2 - j2 \end{aligned}$$

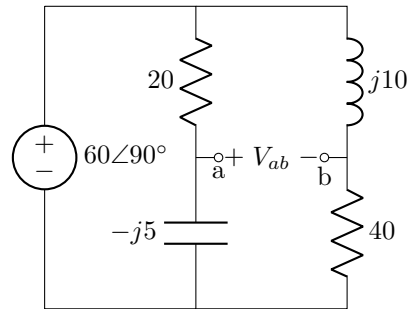
Nodal analysis around supernode:

$$\begin{aligned} \frac{V_1}{2 - j2} + \frac{V_1}{j4} + \frac{V_2}{j2} + \frac{V_2}{1 - j1} &= 0 \\ \frac{V_1}{2 - j2} + \frac{V_1}{j4} + \frac{2 - j2}{j2} + \frac{2 - j2}{1 - j1} &= 0 \end{aligned}$$

$$V_1 = -4 + j4$$

$$\begin{aligned} V_s &= -4 + j4 - 2 + j2 \\ &= -6 + j6 \\ &= 8.485 \angle 135^\circ \text{ V} \end{aligned}$$

Question 9.66

Find $\mathbf{Z}_T$ and $\mathbf{V}_{ab}$:

Memo

 Z_T is the total impedance as seen from the source

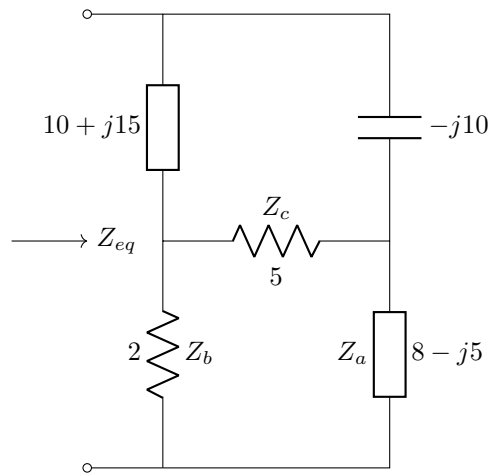
$$\begin{aligned}
 Z_T &= (20 - j5) \parallel (40 + j10) \\
 &= \frac{(20 - j5)(40 + j10)}{(20 - j5) + (40 + j10)} \\
 &= 14.069 - j1.172 \Omega
 \end{aligned}$$

$$\begin{aligned}
 V_a &= 60 \angle 90^\circ \left(\frac{-j5}{20 - j5} \right) \\
 &= 14.1176 + j3.529 \text{ V}
 \end{aligned}$$

$$\begin{aligned}
 V_b &= 60 \angle 90^\circ \left(\frac{40}{40 + j10} \right) \\
 &= 14.1176 + j56.471 \text{ V}
 \end{aligned}$$

$$\begin{aligned}
 V_{ab} &= V_a - V_b \\
 &= 14.1176 + j3.529 - 14.1176 - j56.471 \\
 &= -j52.941 \text{ V} \\
 &= 52.941 \angle -90^\circ \text{ V}
 \end{aligned}$$

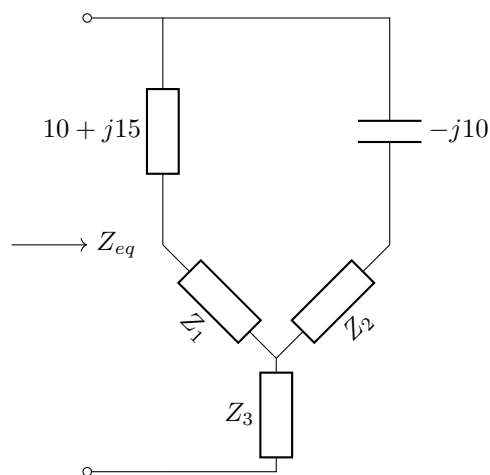
Question 9.70

Find $\mathbf{Z}_{eq}$:

Memo

Use a $\Delta - Y$ Transformation:

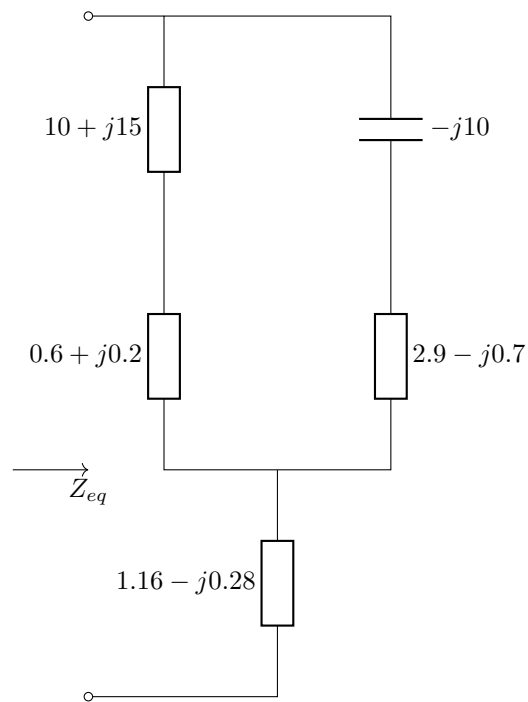
$$Z_a + Z_b + Z_c = 15 - j5$$



$$Z_1 = \frac{Z_b Z_c}{15 - j5} = \frac{(2)(5)}{15 - j5} = 0.6 + j0.2 \, \Omega$$

$$Z_2 = \frac{Z_a Z_c}{15 - j5} = \frac{(8 - j5)(5)}{15 - j5} = 2.9 - j0.7 \, \Omega$$

$$Z_3 = \frac{Z_a Z_b}{15 - j5} = \frac{(8 - j5)(2)}{15 - j5} = 1.16 - j0.28 \, \Omega$$



$$\begin{aligned}
 Z_{eq} &= \{(10 + j15) + (0.6 + j0.2)\} \parallel \{(-j10) + (2.9 - j0.7)\} + (1.16 - j0.28) \\
 &= (10.6 + j15.2) \parallel (2.9 - j10.7) + (1.16 - j0.28) \\
 &= \frac{(10.6 + j15.2)(2.9 - j10.7)}{(10.6 + j15.2) + (2.9 - j10.7)} + (1.16 - j0.28) \\
 &= 12.51 - j9.2 \, \Omega
 \end{aligned}$$